Proposition 8: no linear bound for the local edit distance

Correspondence between the paper and the Lean 4/Mathlib formalisation

Paper. J. Breitung, T. Gervens, M. Grohe. *Graph Similarity Meets Optimal Transport* (file `Graph_Similarity_Meets_Optimal_Transport.pdf` in the repository). Statement, equation, page and line numbers refer to this PDF; `docs/NUMBERING.md` lists the Lean files and declarations of all numbered statements.

The folder `GsimOT/Proposition9_LocalEditLowerBound` formalises Proposition 8 together with the construction of App. D.1 and Lemmas 27 and 28, Theorem 3 (2), and the remarks after Proposition 8 and Theorem 9 that concern G_r, H_r. Shared notions and lemmas (Observation 21, Lemma 22, Lemma 29) come from `GsimOT/Common`; they are listed at the end.

Checking. Run `bash build.sh` in the repository root (see `README.md`). The script builds the library and runs `scripts/CheckAxioms.lean`, which must report only the standard axioms `propext`, `Classical.choice`, `Quot.sound`: no `sorry` and no additional axioms.

Main statement in Lean. `GsimOT.Prop9.proposition9`: for every $r \in \mathbb{N}$ with $2 \le r$ and r even,

$$d_1(G_r, H_r) = r, \qquad d_1^{\mathrm{OT}}(G_r, H_r) = 1, \qquad r^3 \le |V(G_r)| \le 19\, r^3.$$

`GsimOT.Prop9.no_linear_bound`: for every $c \in \mathbb{R}$ there is an even $r \ge 2$ with $\delta_1(G_r, H_r) > c \cdot \delta_1^{\mathrm{OT}}(G_r, H_r)$. `GsimOT.Prop9.no_power_bound`: for all $C \in \mathbb{R}$ and $\alpha > 2/3$ there is an even $r \ge 2$ with $\delta_1(G_r, H_r) > C \cdot (\delta_1^{\mathrm{OT}}(G_r, H_r))^{\alpha}$.

Correspondence: Proposition 8 and its proof

Paper	Page	Statement	Lean declaration(s) and location
Proposition 8	7	For even $r \ge 2$: $d_1(G_r, H_r) = r$, $d_1^{\mathrm{OT}}(G_r, H_r) = 1$, $\lvert V(G_r)\rvert = \Theta(r^3)$	`GsimOT.Prop9.proposition9` `Proposition9_LocalEditLowerBound/Proposition9.lean:53`
Theorem 3 (2)	4	For even $r \ge 2$: $\delta_1(G_r, H_r) = r \cdot \delta_1^{\mathrm{OT}}(G_r, H_r)$	`GsimOT.Prop9.theorem4_item2` `Proposition9_LocalEditLowerBound/Proposition9.lean:61`
after Prop. 8	9	No bound of the form $\delta_1 \le c \cdot \delta_1^{\mathrm{OT}}$	`GsimOT.Prop9.no_linear_bound` `Proposition9_LocalEditLowerBound/Proposition9.lean:69`
after Thm. 9	9	No bound of the form $\delta_1 \le C \cdot (\delta_1^{\mathrm{OT}})^{\alpha}$ with $\alpha > 2/3$	`GsimOT.Prop9.no_power_bound` `Proposition9_LocalEditLowerBound/Proposition9.lean:91`

Paper	Page	Statement	Lean declaration(s) and location
App. D.1	25	$\ell = r^2 + 2r$; core X ($n_X = 1 + r(\ell - 1)$), markers $M = C \cup T \cup B$ ($n_M = 2r^2$)	GsimOT.Prop9.ell Proposition9_LocalEditLowerBound/Construction.lean:37 GsimOT.Prop9.Core Proposition9_LocalEditLowerBound/Construction.lean:49 GsimOT.Prop9.Marker Proposition9_LocalEditLowerBound/Construction.lean:42 GsimOT.Prop9.card_core Proposition9_LocalEditLowerBound/Construction.lean:107 GsimOT.Prop9.card_marker Proposition9_LocalEditLowerBound/Construction.lean:100
App. D.1 (table)	26	Adjacency of core vertices to clocks, tags and bits; patterns χ	GsimOT.Prop9.pat Proposition9_LocalEditLowerBound/Construction.lean:118 GsimOT.Prop9.chiG Proposition9_LocalEditLowerBound/Construction.lean:129 GsimOT.Prop9.chiH Proposition9_LocalEditLowerBound/Construction.lean:134
App. D.1	25–26	G_r^0, H_r^0; $\deg_{H_r^0}(z) = \deg_{G_r^0}(z) + \mathbf{1}[z \in C]$	GsimOT.Prop9.bipGraph Proposition9_LocalEditLowerBound/Construction.lean:182 GsimOT.Prop9.G0 Proposition9_LocalEditLowerBound/Construction.lean:197 GsimOT.Prop9.H0 Proposition9_LocalEditLowerBound/Construction.lean:199 GsimOT.Prop9.deg_H0_marker Proposition9_LocalEditLowerBound/Construction.lean:274

Paper	Page	Statement	Lean declaration(s) and location

App. D.1	26	Lemma 22 applied with markers sorted by degree, $\Delta = n_X$, $\tau = r$, $\delta = 1$	`GsimOT.Prop9.markerEnum`
			`Proposition9_LocalEditLowerBound/LockedGraphs.lean:37`
			`GsimOT.Prop9.markerEnum_mono`
			`Proposition9_LocalEditLowerBound/LockedGraphs.lean:45`
			`GsimOT.Prop9.Gr`
			`Proposition9_LocalEditLowerBound/LockedGraphs.lean:60`
			`GsimOT.Prop9.Hr`
			`Proposition9_LocalEditLowerBound/LockedGraphs.lean:62`
			`GsimOT.Prop9.lock_hyp_deg_G`
			`Proposition9_LocalEditLowerBound/LockedGraphs.lean:88`
			`GsimOT.Prop9.lock_hyp_delta`
			`Proposition9_LocalEditLowerBound/LockedGraphs.lean:91`

App. D.1	26	All vertices of G_r^0, H_r^0 have degree $\le n_X$	`GsimOT.Prop9.deg_bip_le`
			`Proposition9_LocalEditLowerBound/Construction.lean:258`
			`GsimOT.Prop9.card_marker_le_card_core`
			`Proposition9_LocalEditLowerBound/Construction.lean:250`

App. D.1	26	$\chi_{G_r}(u) = 0$, $\chi_{H_r}(u) = \mathbf{1}_C$ (uses r even)	`GsimOT.Prop9.pat_zero`
			`Proposition9_LocalEditLowerBound/Construction.lean:138`
			`GsimOT.Prop9.pat_ell`
			`Proposition9_LocalEditLowerBound/Construction.lean:142`

App. D.1 (Fig. 6)	26	Consecutive patterns differ in (at most) one coordinate	`GsimOT.Prop9.change_clock_iff`
			`Proposition9_LocalEditLowerBound/Patterns.lean:41`
			`GsimOT.Prop9.change_tag_iff`
			`Proposition9_LocalEditLowerBound/Patterns.lean:47`
			`GsimOT.Prop9.change_bit_iff`
			`Proposition9_LocalEditLowerBound/Patterns.lean:57`
			`GsimOT.Prop9.ham_pat_succ_le_one`
			`Proposition9_LocalEditLowerBound/Patterns.lean:67`

Paper	Page	Statement	Lean declaration(s) and location
Proof of Lemma 27	27	Shifts σ_i along branch i	GsimOT.Prop9.node Proposition9_LocalEditLowerBound/LockedGraphs.lean:101 GsimOT.Prop9.shift Proposition9_LocalEditLowerBound/LockedGraphs.lean:119 GsimOT.Prop9.shiftPerm Proposition9_LocalEditLowerBound/LockedGraphs.lean:184 GsimOT.Prop9.sigma Proposition9_LocalEditLowerBound/LockedGraphs.lean:188 GsimOT.Prop9.sigmaExt Proposition9_LocalEditLowerBound/Lemma33_FractionalCost.lean:201
Proof of Lemma 27	27	$Q_r = \frac{1}{r}\sum_i P_{\sigma_i}$ (extended by the identity on K)	GsimOT.Prop9.Qr Proposition9_LocalEditLowerBound/Lemma33_FractionalCost.lean:91 GsimOT.Prop9.Qr_isDS Proposition9_LocalEditLowerBound/Lemma33_FractionalCost.lean:93 GsimOT.Prop9.QrExt Proposition9_LocalEditLowerBound/Lemma33_FractionalCost.lean:187
Proof of Lemma 27	27	Cost of supported core pairs = Hamming distance of patterns	GsimOT.Prop9.localCost_sigma_core Proposition9_LocalEditLowerBound/Lemma33_FractionalCost.lean:38 GsimOT.Prop9.ham_shift_le_one Proposition9_LocalEditLowerBound/Lemma33_FractionalCost.lean:139
Proof of Lemma 27	27	Cost at a marker = mass of changes: clocks 1, tags $2/r$, bits 1	GsimOT.Prop9.localCost_sigma_marker Proposition9_LocalEditLowerBound/Lemma33_FractionalCost.lean:48 GsimOT.Prop9.nchg_clock_le Proposition9_LocalEditLowerBound/Patterns.lean:136 GsimOT.Prop9.nchg_tag_le Proposition9_LocalEditLowerBound/Patterns.lean:144 GsimOT.Prop9.nchg_bit_le Proposition9_LocalEditLowerBound/Patterns.lean:159 GsimOT.Prop9.sum_nchg_le Proposition9_LocalEditLowerBound/Patterns.lean:198
Lemma 27 ($\leq$)	27	$C_1(Q_r) \leq 1$ (computed in (G_r^0, H_r^0) via Lemma 22 (2))	GsimOT.Prop9.costOne_Qr_base_le_one Proposition9_LocalEditLowerBound/Lemma33_FractionalCost.lean:148 GsimOT.Prop9.costOne_Qr_le_one Proposition9_LocalEditLowerBound/Lemma33_FractionalCost.lean:192

Paper	Page	Statement	Lean declaration(s) and location
Lemma 27	27	$d_1^{\mathrm{OT}}(G_r, H_r) = 1$ ($\geq$ by Lemma 29, since $G_r \ncong H_r$)	`GsimOT.Prop9.lemma33` `Proposition9_LocalEditLowerBound/Proposition9.lean:43`

Lemma 28	27	$d_1(G_r, H_r) \geq r$	`GsimOT.Prop9.lemma34`
			`Proposition9_LocalEditLowerBound/Lemma34_BijectionCost.lean:255`
			`GsimOT.Prop9.lemma34_dOne`
			`Proposition9_LocalEditLowerBound/Proposition9.lean:32`

Proof of Lemma 28	27	$C_1(\pi) < r \Rightarrow \pi$ fixes the markers and permutes X (Lemma 22 (1) with $\tau = r$); $L_\pi(x, \pi(x))$ is the Hamming distance of the patterns (used as $\geq$)	`GsimOT.Prop9.lemma34` `Proposition9_LocalEditLowerBound/Lemma34_BijectionCost.lean:255`
Proof of Lemma 28	27	Walk $u, \pi(u), \pi^2(u), \ldots$ and its pattern sequence $x_0, \ldots, x_s$	`GsimOT.Prop9.lemma34` `Proposition9_LocalEditLowerBound/Lemma34_BijectionCost.lean:255`

Proof of Lemma 28	27	Clock level $c(x)$; consecutive clock levels differ by $< r$	`GsimOT.Prop9.clockLevel`
			`Proposition9_LocalEditLowerBound/Patterns.lean:216`
			`GsimOT.Prop9.clockLevel_le_add_ham`
			`Proposition9_LocalEditLowerBound/Patterns.lean:224`

Proof of Lemma 28	27	Middle clock level $\Rightarrow$ all tags of the branch on; patterns on different branches differ in $\geq 2r$ markers (the formalisation uses $\geq r$)	`GsimOT.Prop9.pat_tag_of_middle`
			`Proposition9_LocalEditLowerBound/Patterns.lean:299`
			`GsimOT.Prop9.ham_ge_of_middle_ne`
			`Proposition9_LocalEditLowerBound/Patterns.lean:339`

Proof of Lemma 28	27	$c(x_{\iota_j - 1}) \in ((j-1)r, jr) \Rightarrow$ time in block j, bit value $j \bmod 2$	`GsimOT.Prop9.div_eq_of_clockLevel`
			`Proposition9_LocalEditLowerBound/Patterns.lean:260`
			`GsimOT.Prop9.pat_bit_self`
			`Proposition9_LocalEditLowerBound/Patterns.lean:353`

Proof of Lemma 28	27	α, β, ι_j; the bit sequence alternates r times; $L_\pi(b_i, b_i) \geq r$	`GsimOT.Prop9.walk_bits`
			`Proposition9_LocalEditLowerBound/Lemma34_BijectionCost.lean:65`
			`GsimOT.Prop9.card_changes_step`
			`Proposition9_LocalEditLowerBound/Lemma34_BijectionCost.lean:36`
			`GsimOT.Prop9.lemma34`
			`Proposition9_LocalEditLowerBound/Lemma34_BijectionCost.lean:255`

Paper	Page	Statement	Lean declaration(s) and location
Proof of Prop. 8	27–28	$C_1(\sigma_1) \le r$ (computed in (G_r^0, H_r^0) via Lemma 22 (2))	GsimOT.Prop9.costOne_sigma_base_le Proposition9_LocalEditLowerBound/Lemma33_FractionalCost.lean:170 GsimOT.Prop9.costOneBij_sigmaExt_le Proposition9_LocalEditLowerBound/Lemma33_FractionalCost.lean:204 GsimOT.Prop9.nchg_le Proposition9_LocalEditLowerBound/Patterns.lean:190
Proof of Prop. 8	28	$d_1(G_r, H_r) = r$	GsimOT.Prop9.dOne_Gr_Hr Proposition9_LocalEditLowerBound/Proposition9.lean:36
Proof of Prop. 8	28	$\lvert V(G_r)\rvert = 5r^3 + 10r^2 - r + 4 = \Theta(r^3)$	GsimOT.Prop9.card_Vr Proposition9_LocalEditLowerBound/LockedGraphs.lean:68 GsimOT.Prop9.card_Vr_bounds Proposition9_LocalEditLowerBound/LockedGraphs.lean:76

Shared notions and lemmas used (from GsimOT/Common)

Paper	Page	Statement	Lean declaration(s) and location
Sec. 2	5	Adjacency matrix $A_G \in \{0,1\}^{V\times V}$ (real entries)	GsimOT.adj Common/Basic.lean:41
App. A	14	Degree $\deg_G(v) = \lvert N_G(v)\rvert$; $\sum_{v'} A_G(v,v') = \deg_G(v)$	GsimOT.deg Common/Basic.lean:45 GsimOT.sum_adj_eq_deg Common/Basic.lean:69
App. A, Eq. (6)	14	Discrepancy $D_{G,H}(v,w,v',w') = A_G(v,v') - A_H(w,w')$	GsimOT.disc Common/Basic.lean:78
Sec. 2	5	$\mathrm{DS}(V,W)$: doubly stochastic matrices	GsimOT.IsDS Common/Basic.lean:85
Sec. 2	5	Permutation matrix $P_\pi \in \mathrm{Perm}(V,W)$; $P_\pi \in \mathrm{DS}(V,W)$	GsimOT.permMat Common/Basic.lean:91 GsimOT.permMat_isDS Common/Basic.lean:93

Paper	Page	Statement	Lean declaration(s) and location
Sec. 2	5	Local cost $L_Q(v,w) = \sum_{(v',w')} \lvert D(v,w,v',w')\rvert\, Q(v',w')$	`GsimOT.localCost` `Common/Basic.lean:131`
Sec. 2, Eq. (7)	5, 14	$C_1(Q) = \max_{(v,w)\in\mathrm{supp}(Q)} L_Q(v,w)$	`GsimOT.costOne` `Common/Basic.lean:137` `GsimOT.localCost_le_costOne` `Common/Basic.lean:176` `GsimOT.costOne_le` `Common/Basic.lean:182`
Sec. 2	5	$C_1(\pi) = C_1(P_\pi)$ and $L_{P_\pi}(v,\pi(v)) = \lvert\pi(N_G(v)) \triangle N_H(\pi(v))\rvert$	`GsimOT.costOneBij` `Common/Basic.lean:141` `GsimOT.localCost_permMat_apply` `Common/Basic.lean:207` `GsimOT.costOneBij_le_iff` `Common/Basic.lean:212`
Sec. 1, Sec. 2	1, 5	$d_1(G,H) = \min_{\pi\in\mathrm{Bij}(V,W)} C_1(\pi)$	`GsimOT.dOne` `Common/Basic.lean:144`
Sec. 2	5	$d_1^{\mathrm{OT}}(G,H) = \min_{Q\in\mathrm{DS}(V,W)} C_1(Q)$ (as an infimum over $\mathrm{DS}(V,W)$)	`GsimOT.dOTOne` `Common/Basic.lean:148`
Sec. 1, Sec. 2	1, 5	Normalisations $\delta_1 = n^{-1}d_1$, $\delta_1^{\mathrm{OT}} = n^{-1}d_1^{\mathrm{OT}}$ (scaling by $1/n$)	`GsimOT.deltaOne` `Common/Basic.lean:151` `GsimOT.deltaOTOne` `Common/Basic.lean:154`
Sec. 1, Sec. 3	3, 6	$d_1^{\mathrm{OT}} \le d_1$ (every bijection is a fractional alignment)	`GsimOT.dOTOne_le_dOne` `Common/Basic.lean:264`
Sec. 2, Eq. (3)	5	Birkhoff–von Neumann: $\mathrm{supp}(Q)$ contains the graph of a bijection	`GsimOT.exists_bij_subset_supp` `Common/Birkhoff.lean:20`
Observation 21	20	$L_Q(v,w) \ge \lvert\deg_G(v) - \deg_H(w)\rvert$	`GsimOT.claim26` `Common/Claim26_DegreeBound.lean:21` `GsimOT.claim26_bij` `Common/Claim26_DegreeBound.lean:40`

Paper	Page	Statement	Lean declaration(s) and location
Lemma 22 (construction)	21	Adding the clique K ($m = 2\Delta + 2\tau + k(\tau+\delta)+1$) and the staircase	`GsimOT.lockSize` Common/Lemma27_LockingMarkers.lean:36 `GsimOT.stair` Common/Lemma27_LockingMarkers.lean:40 `GsimOT.lockGraph` Common/Lemma27_LockingMarkers.lean:56
Lemma 22 (1)	21	$C_1(\pi) < \tau \Rightarrow \pi$ fixes all markers and $\pi(V) = V$	`GsimOT.lemma27_bij` Common/Lemma27_LockingMarkers.lean:161
Lemma 22 (2)	21	$Q(z_j, z_j) = 1 \Rightarrow C_1(Q')$ w.r.t. (G', H') equals $C_1(Q)$ w.r.t. (G, H)	`GsimOT.extendId` Common/Lemma27_LockingMarkers.lean:260 `GsimOT.lemma27_cost` Common/Lemma27_LockingMarkers.lean:351
App. D.2	28	Pseudometric $\rho_H(w, w') = \lvert N_H(w) \triangle N_H(w') \rvert$ and the triangle inequality	`GsimOT.rho` Common/Lemma35_Integrality.lean:25 `GsimOT.rho_triangle` Common/Lemma35_Integrality.lean:36
Lemma 29 (first part)	28	$Q(v, w) > 0 \Rightarrow \rho_H(\sigma(v), w) \le 2L$	`GsimOT.rho_le_localCost_add` Common/Lemma35_Integrality.lean:43 `GsimOT.lemma35_rho_le` Common/Lemma35_Integrality.lean:110
Lemma 29 (second part)	28	$L < 1 \Rightarrow \sigma$ is an isomorphism	`GsimOT.rho_eq_zero_of_deg_eq_of_lt_two` Common/Lemma35_Integrality.lean:79 `GsimOT.lemma35_iso` Common/Lemma35_Integrality.lean:120 `GsimOT.lemma35_dOne_eq_zero` Common/Lemma35_Integrality.lean:162
Lemma 29 ("in particular")	28	$G \not\cong H \Rightarrow d_1^{\mathrm{OT}}(G, H) \ge 1$	`GsimOT.lemma35_one_le_dOTOne` Common/Lemma35_Integrality.lean:176

Notes on the formalisation

1. Proposition 8 states $\Theta(r^3)$ vertices. The formalisation proves $r^3 \le |V(G_r)| \le 19r^3$ and, more precisely, $|V(G_r)| = 5r^3 + 10r^2 - r + 4$ (`card_Vr`).

2. Indices are 0-based in Lean: `Marker.clock j h` is $c_{j+1,h+1}$, `Marker.tag i h` is $t_{i+1,h+1}$, `Marker.bit i` is b_{i+1} and `Core.v i k` is $v_{i+1,k+1}$.

3. Adjacency is defined through the pattern `pat` $r\,i\,t$ of branch i at time $t \in [0, \ell]$, which is exactly the table of App. D.1 (p. 26, l. 800) with k replaced by t. Thus $\chi(v_{i,k}) = \mathtt{pat}\,i\,k$, $\chi_{G_r}(u) = \mathtt{pat}\,i\,0 = 0$ and $\chi_{H_r}(u) = \mathtt{pat}\,i\,\ell = \mathbf{1}_C$.

4. The proof of Lemma 27 obtains $d_1^{\mathrm{OT}}(G_r, H_r) \geq 1$ from Lemma 29 since $G_r \ncong H_r$ (p. 27, l. 837). The formalisation derives $G_r \ncong H_r$ from Lemma 28: every bijection has cost $\geq r > 0$, while an isomorphism would have cost 0. (The proof idea, p. 8, l. 282–283, obtains the equality from Observation 21 instead: u is isolated in G_r but H_r has no isolated vertex.)

5. In the proof of Lemma 27 (p. 27, l. 836–837) the paper computes the local costs at clock markers, tags and bits as exactly 1, $2/r$ and 1. The formalisation proves the upper bounds 1, $2/r$ and 1, which are all that is needed for $C_1(Q_r) \leq 1$.

6. Proof of Proposition 8: for $d_1(G_r, H_r) \leq r$ the formalisation shows that *every* local cost of σ_1 is at most r (core pairs ≤ 1, clocks ≤ 1, tags ≤ 2, bits $\leq r$), which is the content of "its local costs can be computed ... as in the proof of Lemma 27".

7. Proof of Lemma 28: the walk is formalised via the permutation P of the core induced by π and its orbit $u, P(u), \ldots, P^s(u) = u$, with s the least positive period. The pattern sequence is $x_\iota = \chi_{G_r}(P^\iota(u))$ for $\iota < s$ and $x_s = \chi_{H_r}(u)$, as in the paper. The indices α, β, ι_j are defined exactly as in the paper (via `Nat.find` / `Nat.findGreatest`). For patterns on different branches the formalisation only uses Hamming distance $\geq r$ (the paper notes $\geq 2r$), which suffices. At a core vertex x it only uses that $L_\pi(x, \pi(x))$ is at least the Hamming distance of the patterns (the paper: equal to it, p. 27, l. 840–841).

8. The remark after Theorem 9 (p. 9, l. 318–320): since $\delta_1(G_r, H_r) = r/n$, $\delta_1^{\mathrm{OT}}(G_r, H_r) = 1/n$ and $n = |V(G_r)| \leq 19r^3$, no bound $\delta_1 \leq C(\delta_1^{\mathrm{OT}})^\alpha$ with $\alpha > 2/3$ holds (`no_power_bound`). The exact asymptotics $\Theta(r^{-3})$, $\Theta(r^{-2})$ are not stated separately.